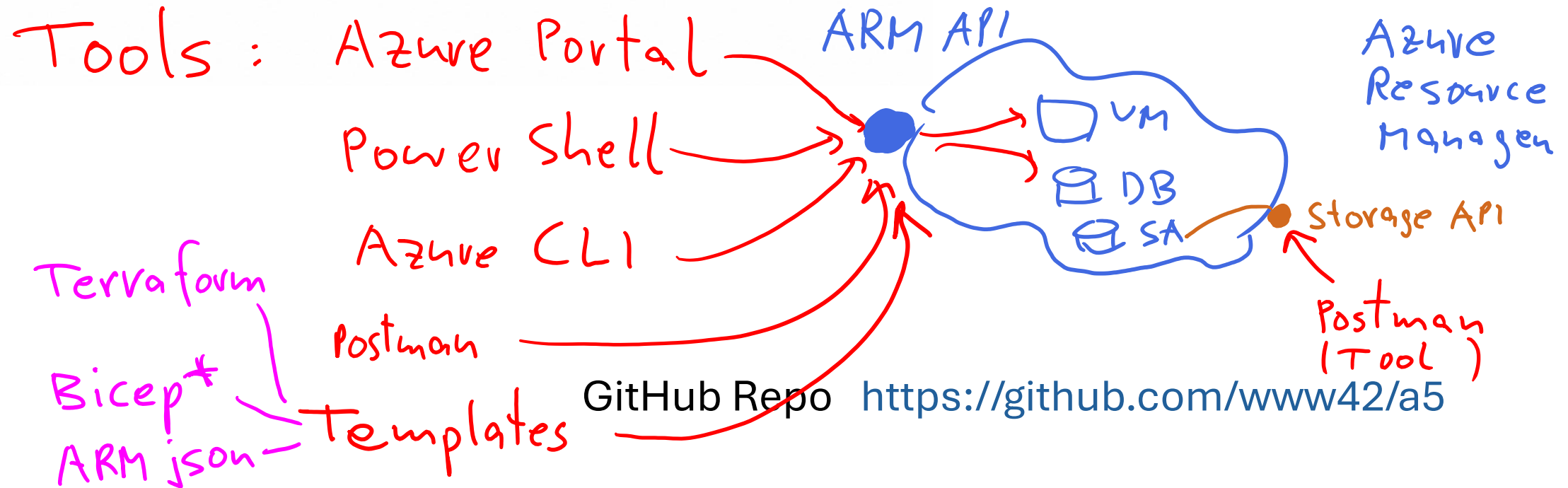




29.06.2026
Tag 6

Von KI erstellt, von Menschen kontrolliert.



AZ-104

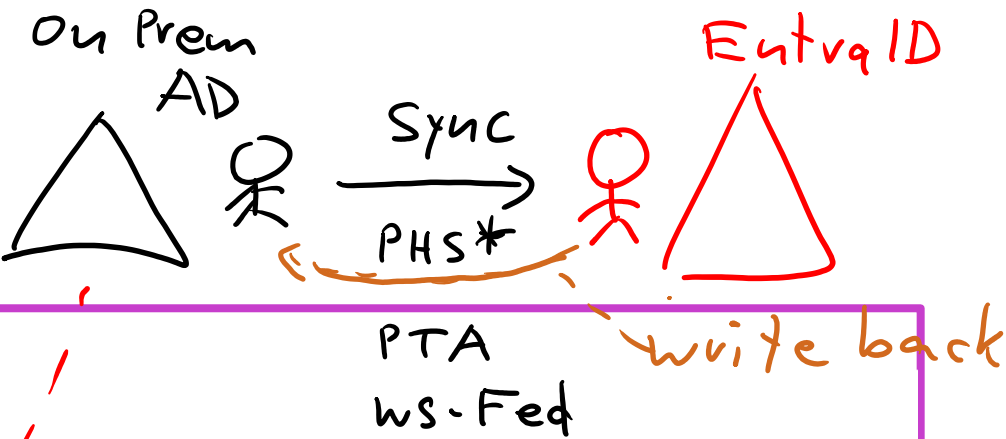
Learning Paths:

- Manage identities and governance in Azure ✓
- Prerequisites for Azure administrators – CloudShell and templates *Tools*
- Implement and manage storage in Azure
- Configure and manage virtual networks for Azure administrators
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

What happened on Friday



Enabling SSPR



Password reset | Properties

✕ Diagnose and solve problems

Save Discard

Manage

- Properties
- Authentication methods
- Registration
- Notifications
- Customization
- On-premises integration
- Administrator Policy

Self service password reset enabled ⓘ

None **Selected** All

Select group ⓘ

00000000-0000-0000-0000-000000000000

These settings only apply to end users in your organization. Admins are always enabled for self-service password reset and are required to use two authentication methods to reset their password. {0}.

PTA = Pass Through Auth.
WS-Fed = Web Services for Federation

WS-Man = Web Services for Management
dir wsman: = WinRM

AZ-104

Lab 02b - Manage Governance via Azure Policy

~~X~~
Ging nicht am Freitag :)



Hente Nachmittag:

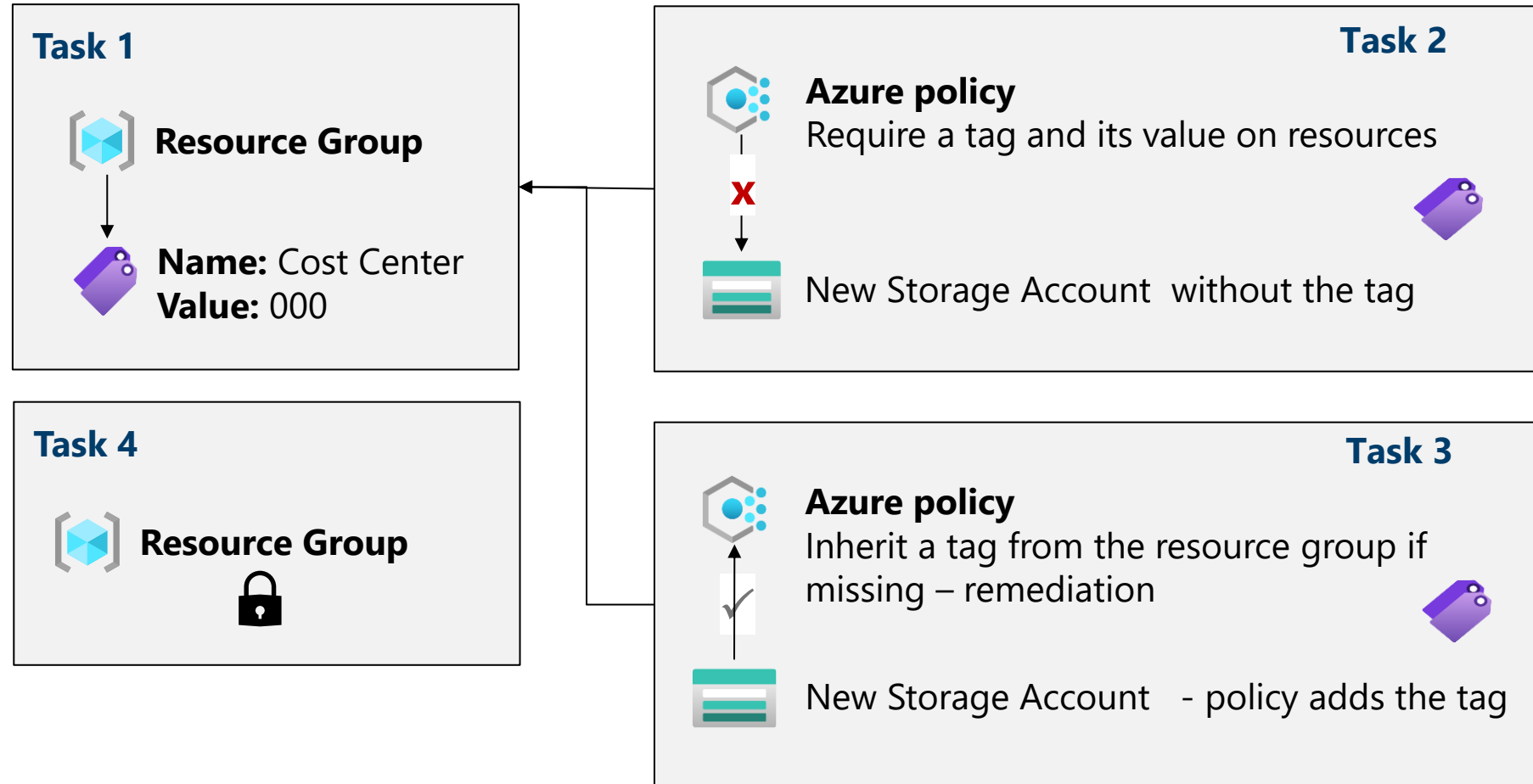
1. Lab 08 (AZ-104) VM Portal / Powershell / CLI

2. TODO Liste Windows 11 AS

3. Lab 026 (AZ-104) Azure Policy

Pause bis 13:00 Uhr

Lab 02b – Architecture diagram



AZ-104

Prerequisites for Azure administrators – CloudShell and templates



Configure Azure Resources with the Azure Portal, PowerShell, and the CLI

Azure CLI
Tool az (Binary)



Review Resource Manager Benefits

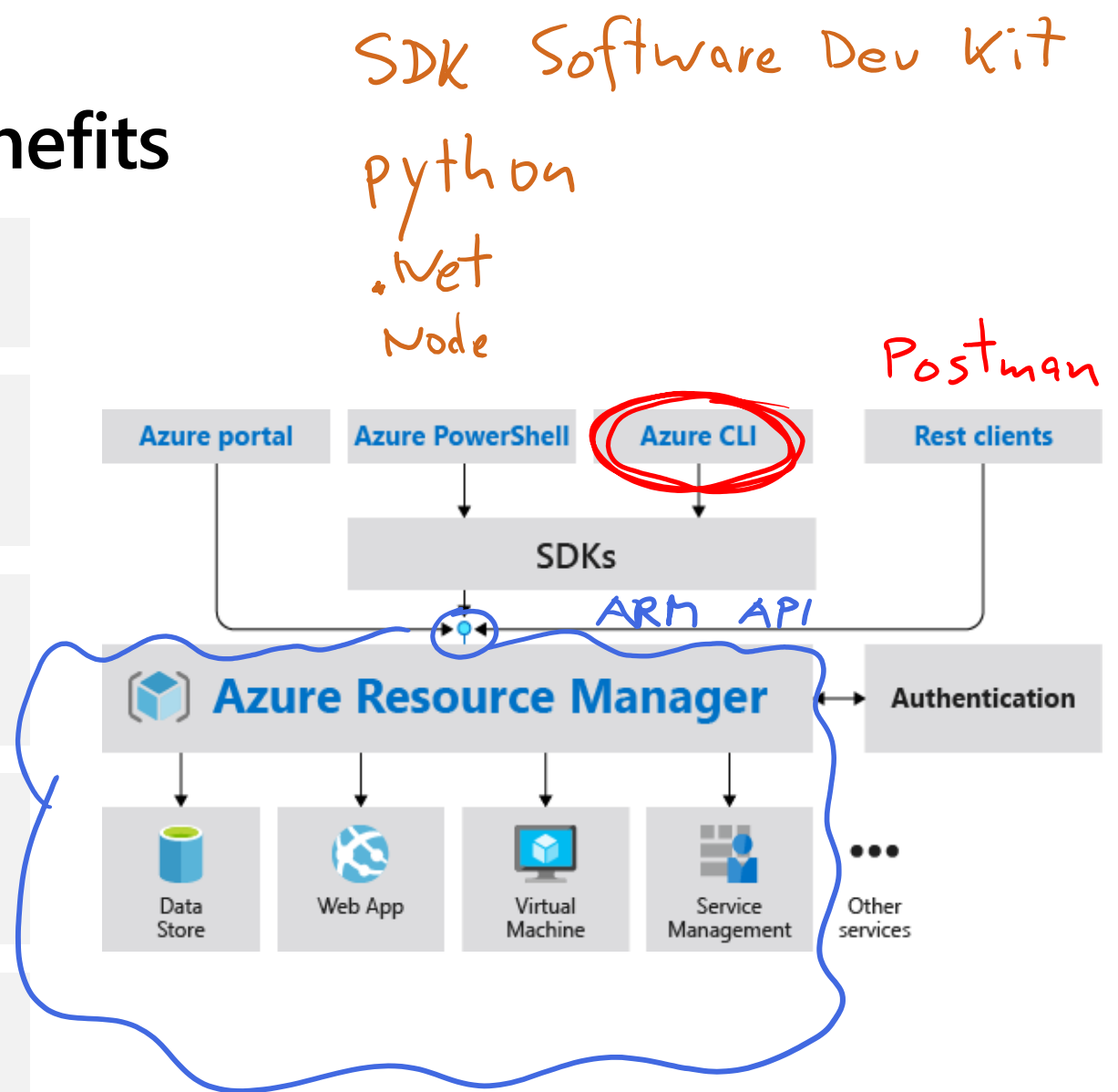
Provides a consistent management layer

Enables you to work with the resources in your solution as a group

Deploy, update, or delete in a single, coordinated operation

Provides security, auditing, and tagging features

Choose the tools and APIs that work best for you



Use Azure Cloud Shell

= Docker Container in Azure (ACI)

Interactive, browser-accessible shell

Offers either Bash or PowerShell

Is temporary and provided on a per-session, per-user basis

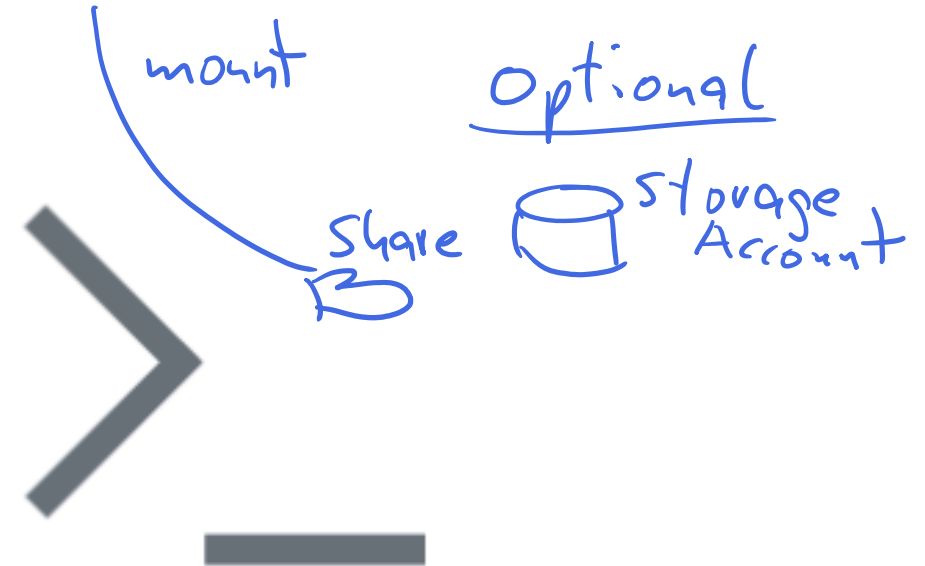
Optional
Requires a resource group, storage account, and Azure File share

Authenticates automatically

Integrated graphical text editor *code, nano, vi*

Is assigned one machine per user account

Times out after 20 minutes



Welcome to Azure Cloud Shell



2x Container OE

Hardware Azure
ich bin schon authentifiziert

Linux Cloud Shell

PS > Get-AzVm

API Call

GET
Access Token

ARM
API

vm1
vm2

vm1
vm2

API Response

ich muß mich erst anmelden an Azure

Windows 11
Hardware

Local Shell

PS > connect-AzAccount...
Get-AzVm

API Call
GET
Access Token

API Response

Other Administrator Tools

Portal oder Scripting?

PowerShell

Bash

Cloud Shell oder Local Shell

PowerShell

Bash zsh

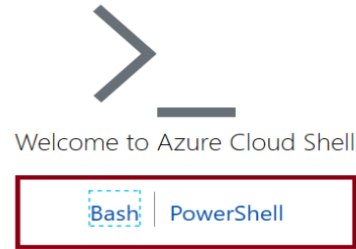
vs code

Azure Portal



- View and manage resources
- Visual interface
- Unified hub – training and documentation
- Personalize your experience
- Mobile app
- Access the Cloud Shell
- One-off creation scenarios

Azure Cloud Shell



- Interactive and browser-accessible with file storage
- Offers Bash or PowerShell
- Authenticates automatically
- Provided on a per-session and per-user basis
- Temporary - times out after 20 minutes

Azure PowerShell and CLI

```
az vm restart -g  
MyResourceGroup -n MyVm
```

- Command line programs
- Interactive and scripting modes
- Cross-platform
- Good for repeatable deployments
- Familiar coding experience

Microsoft PowerShell



Leibniz

- 2002 Jeffrey Snover Monad Manifesto
- 2006 PowerShell 1.0
- 2016 Open-source, cross-platform PowerShell 6
- 2020 PowerShell 7

20 Jahre PowerShell

100 Jahre Quantenmechanik

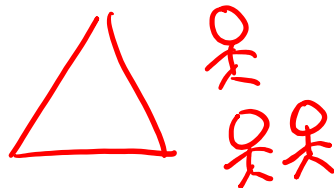
Ignite Atlanta 2016



PS Gallery



Vier PowerShell Module für Entra ID



MSONline ~~×~~ } deprecated
AzureAD ~~×~~ }
Microsoft.Graph ✓ ✓
Microsoft.Entra ✓ ✓

Wann was verwenden?

Szenario	Empfehlung
Altes Skript läuft noch mit MSONline/AzureAD	Kurzfristig lassen, mittelfristig migrieren
Neue Skripte, maximale Kontrolle & API-Nähe	Microsoft.Graph
Neue Skripte, <u>lesbar</u> & AzureAD-ähnlich	Microsoft.Entra
CI/CD Pipelines, komplexe Automation	Microsoft.Graph + Invoke-MgGraphRequest
Schnelle Ad-hoc-Aufgaben im Terminal	Microsoft.Entra

Praktischer Tipp

Microsoft.Entra ist letztlich nur syntaktischer Zucker über Microsoft.Graph — du kannst beide im selben Skript mischen. Wenn ein Microsoft.Entra-Cmdlet nicht das kann was du brauchst, fällst du einfach auf Invoke-MgGraphRequest zurück.



Ein PowerShell Module für Azure

Az 



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6,192,981

Downloads of 16.0.0

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6/2/2026

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Az 16.0.0

Microsoft Azure PowerShell - Cmdlets to manage resources in Azure. This module is compatible with PowerShell and Windows PowerShell.

For more information about the Az module, please visit the following: <https://learn.microsoft.com/powershell/azure/>

Minimum PowerShell version
5.1

Installation Options

Install Module

Manual Download

Copy and Paste the following command to install this package using PowerShellGet [More Info](#)

```
PS> Install-Module -Name Az
```

abhängige module

Labs

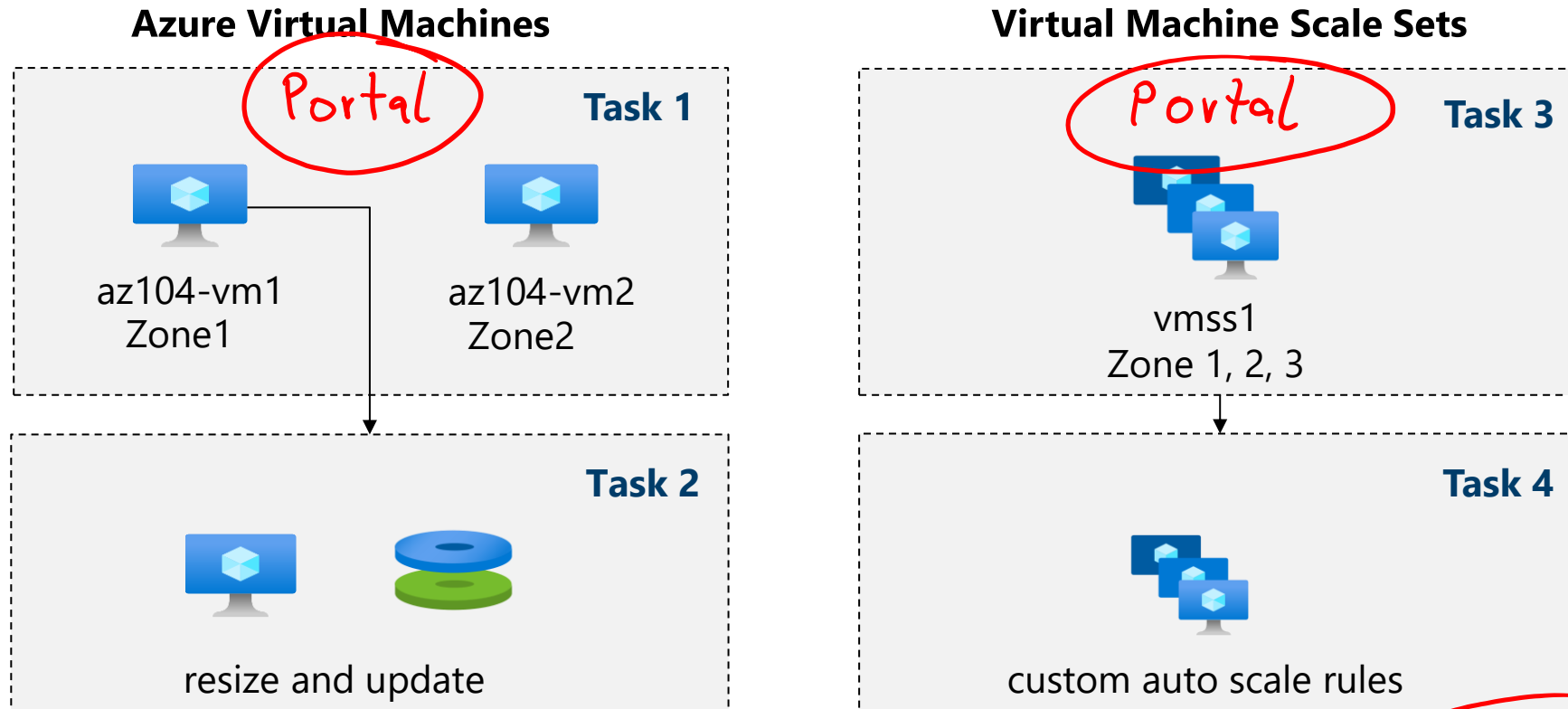


AZ-104

Lab 08 – Manage Virtual Machines



Lab 08 – Architecture diagram



Task 5: Create a virtual machine using Azure PowerShell (option 1)

Task 6: Create a virtual machine using the CLI (option 2)

PowerShell
Azure CLI } Cloud shell

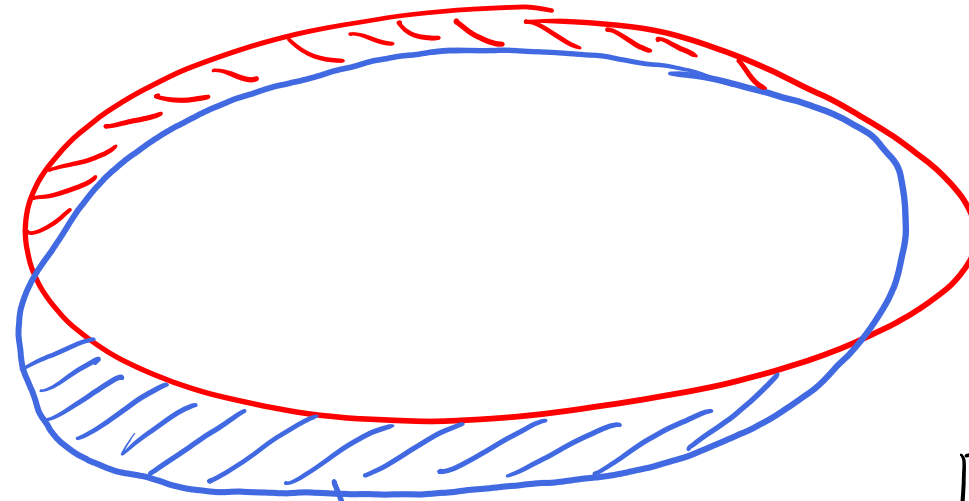
End of presentation



geht nur in PS

Funktionsumfang

PowerShell
modules
.Net Objects



geht nur in CLI

Azure CLI
Groups

json Objects

az vm list -o table
Binary Group
Output Format
(statt json)

	Power shell shell	Bash shell
Power shell cmds	✓	✗
Azure CLI az....	✓	✓